

Quadratic Exponential Feedback after Reversion

Cancellation, finite-core universality,
and sharp Borel growth

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Abstract

Let $\widehat{U}(q) = \sum_{j \geq 1} q^j \exp(a j^2 \widehat{U}(q))$, where $a > 0$, and write $\widehat{V} = \widehat{U}^{(-1)} = \sum v_n z^n$. We prove the quadratic inverse equivalent proposed in the ProveIt finite-core feedback manuscript:

$$v_n \sim -S_n \exp(-(1 + 1/a)r_n), \quad r_n(1 + r_n)e^{2r_n} = an,$$

where

$$S_n = \frac{\exp\left(\frac{nr_n}{1 + r_n}(1 + 2r_n)\right)}{\sqrt{1 + 4r_n + 2r_n^2}}.$$

The missing cancellation estimate is supplied by inverting the defining kernel first, grouping the resulting exact tree expansion by large actions, and merging multiple large actions in a positive majorant. The argument is not an absolute estimate on the powers in the reversion formula for $\widehat{U}$ itself.

The result extends to every finite nonnegative perturbation of the weights and slopes, and to eventually affine-quadratic slopes. Only $\mu = \lambda_1 + w_2$ from the finite core enters the leading inverse factor. We also determine the leading relative error of every fixed core with at least two actions: the first omitted nonzero action ℓ contributes $w_\ell j_n t_n^{\ell-1}$, with $j_n = n/(1 + r_n)$ and $t_n = r_n(1 + r_n)/(an)$. A borderline slope perturbation $\kappa j / \log j$ changes the leading constant by $e^{\kappa/(2a)}$. Finally, the factorial-normalized inverse Borel transform is entire and has a sharp double-exponential positive-ray equivalent, including its prefactor. These are conventional mathematical proofs, accompanied by reproducible exact finite checks; no Lean verification or literature-wide priority claim is made.

Scope and status. The main result resolves a specific conjecture in the inspected repository snapshot, not a general conjecture about all transseries. The new proofs are self-contained apart from classical formal Lagrange inversion, the analytic implicit function theorem, and Stirling's formula. One separately labeled comparison with the forward Borel transform additionally uses the forward coefficient theorem in the repository manuscript. Global originality and publication priority have not been established.

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1 The precise repository question

1.1 What is being proved

The ProveIt transseries project separates its Lean developments from its research manuscripts [1]. Its article *Finite-Core Universality and Sharp Large Order for Countable Exponential-Feedback Transseries* [2] proves forward coefficient asymptotics for the kernel

$$\widehat{U}(q) = \sum_{j \geq 1} q^j e^{aj^p \widehat{U}(q)}, \quad a > 0, \quad p > 1.$$

For $p > 2$ it also proves a reversion asymptotic. At $p = 2$, however, the parameter controlling its absolute convolution argument no longer tends to zero. The manuscript explicitly leaves the inverse equivalent as `Conjecture conj:quadratic-inverse`, in the subsection “Quadratic and subquadratic inverse asymptotics.”

In that notation the conjecture is exactly

$$v_n \sim -S_n \exp(-(1 + 1/a)r_n). \quad (1.1)$$

The manuscript derives an exact one-tail inverse-response identity but does not prove that the response exhausts the signed inverse coefficient. That missing implication, rather than the response identity or Lagrange inversion itself, is the principal issue settled here.

The inspected source is pinned at commit

0c973d8f5e7ef4550f4df1bf486d890037fd9f14,

with TeX blob c2b6faf623003111e9d9e6bbb162a67429f3ce42. The source directory is

Analysis/Transseries/docs/series-and-transseries/Finite_Core_Universality_Exponential_Feedback/.

The source ranges inspected include its definitions, principal theorems, formal coefficient identity, reversion discussion, diagnostics, and precise conjecture. This is not an audit of every report, branch, or incoming archive in the repository.

1.2 Why an inverse-first argument is different

A coefficient of a formal inverse is an alternating sum of convolutions. In the quadratic model the individual endpoint contributions have a collective exponential factor on the scale $\exp(C \log^2 n)$. Discarding all but the first one is not legitimate. Even obtaining a relative $o(1)$ estimate for the forward coefficient is not, by itself, enough to control the substantially smaller inverse coefficient.

We instead use the defining equation at the inverse value:

$$z = \sum_{j \geq 1} w_j \widehat{V}(z)^j e^{\lambda_j z}. \quad (1.2)$$

The exponent now contains the independent variable z , not a divergent unknown series. Small actions can therefore be summed in an analytic core. The remaining cancellation penalty is only a fixed power of n , which a sufficiently large fixed core can dominate. Once this coarse signed reduction has been proved, comparison of two analytic cores gives the *sharp* small-core error.

1.3 Relation to established work

Formal Lagrange inversion and its tree interpretations are classical; see Gessel's survey [3]. Bender's work [4] establishes asymptotic transfer for suitable rapidly growing coefficients under formal operations. Bender and Richmond [5] also treat asymptotic expansions under compositional inversion and analytic substitution. Thus asymptotic reversion itself is an established subject. The contribution here is the particular signed large-action estimate, its quadratic equivalent, and the sharp finite-core defect, not the general idea of transferring asymptotics through inversion.

Borinsky [6] develops an algebra and asymptotic derivation for fixed factorial scales $\alpha^{n+\beta}\Gamma(n+\beta)$. There is an important distinction here: our coefficients are smaller than every such fixed positive factorial scale, although their ordinary generating series diverges. They belong to the zero-asymptotic part of those scales rather than furnishing a nonzero leading coefficient there. A moving saddle is needed to recover their nonzero leading term. This is not a claim that Borinsky's algebra excludes the series.

The transseries interpretation is simply the substitution $q = e^{-x}$ in an ordinary formal series. The support remains $\{e^{-nx} : n \geq 1\}$. The new issue is large-order coefficient information and its behavior under reversion, not a new monomial field. Edgar [7] supplies broader background on the formal and analytic distinctions.

2 Definitions and principal results

2.1 An eventually quadratic family

Assume throughout that

$$a > 0, \quad w_1 = 1, \quad w_j, \lambda_j \geq 0, \quad w_j = 1, \quad \lambda_j = aj^2 + dj + e \quad (j > J_0), \quad (2.1)$$

where J_0 is fixed and $d, e \in \mathbb{R}$ are fixed. In particular the eventual quadratic expression is nonnegative. All finitely many exceptional weights and slopes are finite. Set

$$b = \lambda_1, \quad \mu = b + w_2. \quad (2.2)$$

There is a unique formal solution

$$\widehat{U}(q) = \sum_{j \geq 1} w_j q^j e^{\lambda_j \widehat{U}(q)} = q + O(q^2), \quad \widehat{V} = \widehat{U}^{(-1)} = \sum_{n \geq 1} v_n z^n. \quad (2.3)$$

Existence and the inverse equation are proved in Section 3. The *pure model* has $w_j = 1$, $\lambda_j = aj^2$ for every j ; then $d = e = 0$ and $\mu = a + 1$.

For $n > 0$ define r_n by the strictly increasing scalar equation

$$r_n(1 + r_n)e^{2r_n} = an, \quad j_n = \frac{n}{1 + r_n}, \quad k_n = \frac{nr_n}{1 + r_n}, \quad t_n = \frac{k_n}{aj_n^2} = \frac{r_n(1 + r_n)}{an}. \quad (2.4)$$

The variables j_n, k_n are real saddle coordinates, not integer indices. Put

$$\Delta_n = 1 + 4r_n + 2r_n^2, \quad S_n = \frac{e^{k_n(1+2r_n)}}{\sqrt{\Delta_n}}, \quad L_n = S_n e^{(d-\mu)r_n/a}. \quad (2.5)$$

All limits are taken with the model parameters fixed.

Theorem 2.1 (Inverse universality at the quadratic boundary). *For every model satisfying (2.1),*

$$v_n \sim -L_n = -S_n \exp\left(\frac{d-\mu}{a}r_n\right). \quad (2.6)$$

In particular the coefficients are eventually negative. Moreover,

$$\frac{v_n}{v_{n-1}} \sim e^{2r_n} = \frac{an}{r_n(1+r_n)} \sim \frac{4an}{(\log n)^2}. \quad (2.7)$$

For the pure model this proves (1.1).

The constant term e in the eventual quadratic slope does not enter the leading equivalent. Neither do the individual finite slopes λ_j for $j \geq 2$. Their effects appear at smaller orders. Finite changes to λ_1 or w_2 are exceptional because they change μ .

2.2 Finite cores and their sharp defects

For each fixed $M \geq 1$ let Q_M be the analytic solution near zero of

$$Q_M = ze^{-bz} - \sum_{j=2}^M w_j Q_M^j e^{(\lambda_j - b)z}. \quad (2.8)$$

An empty sum is zero. Define

$$D_M(z) = \left(1 + \sum_{j=2}^M j w_j Q_M(z)^{j-1} e^{(\lambda_j - b)z} \right)^{-1}, \quad (2.9)$$

$$T_M(z) = D_M(z) \sum_{j>M} w_j Q_M(z)^j e^{(\lambda_j - b)z}, \quad A_{n,M} = [z^n] T_M(z). \quad (2.10)$$

The core and D_M are analytic. The tail T_M is only formal. Each of its coefficients is a finite sum, so these definitions do not evaluate a divergent kernel at a positive argument.

Theorem 2.2 (Sharp first-omitted-action law). *Fix $M \geq 2$, and let*

$$\ell = \min\{j > M : w_j > 0\}.$$

This integer exists under (2.1). Then $A_{n,M} > 0$ eventually, $A_{n,M} \sim L_n$, and

$$v_n = -A_{n,M} \left(1 - w_\ell j_n t_n^{\ell-1} + o(j_n t_n^{\ell-1}) \right). \quad (2.11)$$

In the pure model, $\ell = M + 1$ and the relative defect is

$$\frac{v_n + A_{n,M}}{A_{n,M}} \sim j_n t_n^M = \frac{r_n^M (1 + r_n)^{M-1}}{a^M n^{M-1}}. \quad (2.12)$$

Thus $-A_{n,M}$ is eventually more negative than v_n .

The estimate is substantially stronger than a one-tail equivalence. It identifies the coefficient, sign, power of n , and logarithmic scale of the first omitted interaction. The proof uses a much larger auxiliary core, but the conclusion concerns the fixed core M itself.

2.3 Sharp growth of the inverse Borel transform

Our Borel normalization is

$$\mathcal{B}_V(t) = \sum_{n \geq 1} \frac{v_n}{n!} t^n. \quad (2.13)$$

Some conventions instead divide v_n by $(n-1)!$ and lower the exponent of t by one; that is the derivative of (2.13) and is not our convention.

Theorem 2.3 (Entire Borel transform with a sharp positive-ray equivalent). *The function (2.13) is entire. For $t \rightarrow +\infty$, set*

$$R = R(t) = \frac{\sqrt{1+4at} - 1}{2}, \quad R(1+R) = at. \quad (2.14)$$

Then

$$-B_V(t) \sim \frac{\exp\left(\frac{Re^{2R}}{a} + \frac{d-\mu}{a}R\right)}{\sqrt{2R+1}}. \quad (2.15)$$

In particular it has infinite entire-function order and is not of exponential type on the positive ray. The ordinary positive-direction Borel–Laplace integral diverges for every positive Laplace parameter.

3 Exact formal algebra before asymptotics

Lemma 3.1 (Formal existence and the inverse-first equation). *The solution (2.3) exists uniquely and has linear coefficient one. Its compositional inverse is the unique element of $z\mathbb{R}[[z]]$ satisfying*

$$\widehat{V}(z) = ze^{-bz} - \sum_{j \geq 2} w_j \widehat{V}(z)^j e^{(\lambda_j - b)z}. \quad (3.1)$$

In particular $v_2 = -\mu$.

Proof. The coefficient of q^n in the right side of the defining equation for $\widehat{U}$ uses only its coefficients of degrees below n . This is a triangular recursion, and the coefficient of q is $w_1 = 1$. A formal series with zero constant coefficient and unit linear coefficient has a unique compositional inverse. Substitute $q = \widehat{V}(z)$ in the identity for $\widehat{U}$ and use $\widehat{U}(\widehat{V}(z)) = z$. Separating its first summand gives (3.1). Conversely that equation has its own triangular recursion and hence determines the same inverse. At degree two it gives $v_2 = -b - w_2$. $\square$

3.1 A multiplicity expansion

For a finitely supported vector $\mathbf{c} = (c_1, c_2, \dots)$ of nonnegative integers, use excess indices $k = j - 1$ and put

$$K = \sum_{k \geq 1} kc_k, \quad s = \sum_{k \geq 1} c_k, \quad C(\mathbf{c}) = \frac{(K+s)!}{(K+1)! \prod_k c_k!}, \quad W(\mathbf{c}) = \prod_k w_{k+1}^{c_k}. \quad (3.2)$$

For $K \geq 1$ set

$$E_-(\mathbf{c}) = \sum_{k \geq 1} \lambda_{k+1} c_k - b(K+s+1). \quad (3.3)$$

This real number need not be positive in the perturbed family.

Lemma 3.2 (Exact signed tree expansion). *As a coefficientwise well-defined formal identity,*

$$\widehat{V}(z) = ze^{-bz} + \sum_{\mathbf{c}: K \geq 1} (-1)^s C(\mathbf{c}) W(\mathbf{c}) z^{K+1} e^{E_-(\mathbf{c})z}. \quad (3.4)$$

Consequently, for $n \geq 1$,

$$v_n = \frac{(-b)^{n-1}}{(n-1)!} + \sum_{K=1}^{n-1} \sum_{\sum kc_k=K} (-1)^s C(\mathbf{c}) W(\mathbf{c}) \frac{E_-(\mathbf{c})^{n-K-1}}{(n-K-1)!}. \quad (3.5)$$

A zeroth power is interpreted as one, including when its base is zero.

Proof. Temporarily regard z as a parameter, and set

$$x = ze^{-bz}, \quad H_z(q) = \sum_{j \geq 2} w_j q^j e^{(\lambda_j - b)z}.$$

Formal Lagrange inversion for $q = x - H_z(q)$ gives

$$q = x + \sum_{s \geq 1} \frac{(-1)^s}{s!} \left(\frac{\partial}{\partial x} \right)^{s-1} H_z(x)^s.$$

In a term with excess multiplicities $\mathbf{c}$, the total power of x before differentiation is $K + s$. The ordered choices of the s actions contribute $s! / \prod c_k!$. Differentiation therefore gives $C(\mathbf{c})x^{K+1}$. After substituting $x = ze^{-bz}$ the exponent is

$$\sum_k (\lambda_{k+1} - b)c_k - b(K + 1) = E_-(\mathbf{c}).$$

This proves (3.4). At degree n only $K \leq n - 1$ can occur, and there are finitely many partitions of each K . Thus all substitutions and groupings are formal and finite at every coefficient. Extracting the coefficient proves (3.5). $\square$

3.2 Marking large actions

Call an action *large relative to M* when $j > M$, equivalently when its excess k is at least M . Multiply each such primitive weight by a marker h , and let $V_h(z)$ solve (3.1) with these marked weights. Then

$$V_h = Q_M - hT_M + \sum_{r \geq 2} h^r V_{M,r}. \quad (3.6)$$

At a fixed z -degree this is a polynomial in h , since a term with r large actions has $K \geq rM$.

Lemma 3.3 (Core and one-tail identities). *The coefficient of h^0 in (3.6) is the analytic core (2.8), and the coefficient of h is $-T_M$ as in (2.10). For $M \geq 2$,*

$$Q_M(z) = z - \mu z^2 + O(z^3), \quad D_M(z) = 1 + O(z). \quad (3.7)$$

Proof. At $(z, q) = (0, 0)$ the derivative with respect to q of the left side of (2.8), after moving all terms to the left, is one. The analytic implicit function theorem gives a unique analytic core. The denominator of D_M also equals one at zero. Differentiate the marked equation with respect to h at $h = 0$. This gives

$$\left(1 + \sum_{j=2}^M j w_j Q_M^{j-1} e^{(\lambda_j - b)z} \right) \partial_h V_h|_{h=0} = - \sum_{j > M} w_j Q_M^j e^{(\lambda_j - b)z}.$$

This is the claimed response. The degree-two calculation is the one in Lemma 3.1, since action two is included when $M \geq 2$. $\square$

Remark 3.4. The response identity is consistent with the one already present in [2]. In its notation, the factor $Q'R(Q)$ equals the reciprocal partial derivative of the finite forward kernel with respect to its first variable. After removing e^{bz} , this is exactly our D_M and exponential factor. The new step below is a bound on all terms of marker degree at least two.

4 The bare saddle and analytic twists

4.1 A positive reference sequence

Define

$$B_n = \sum_{j=1}^n \frac{(aj^2)^{n-j}}{(n-j)!}, \quad F_n(x) = (n-x) \left(1 + \log \frac{ax^2}{n-x} \right) \quad (0 < x < n). \quad (4.1)$$

The summand with $j = n$ is one.

Lemma 4.1 (Bare saddle and localization). *With the notation (2.4)–(2.5),*

$$B_n \sim S_n. \quad (4.2)$$

The saddle is $x = j_n$ and its Gaussian variance is

$$\sigma_n^2 = \frac{k_n}{\Delta_n} \sim \frac{2n}{(\log n)^2}. \quad (4.3)$$

For any fixed $H \geq 1$ and fixed $B \geq 0$, there is $c > 0$ such that

$$\sum_{\substack{1 \leq K \leq n-1 \\ K \notin I_n}} H^K \frac{(aK^2 + BK + B)^{n-K-1}}{(n-K-1)!} = O(S_n e^{-cn}), \quad I_n = \left[\frac{n}{\log n}, \frac{4n}{\log n} \right]. \quad (4.4)$$

The values of c and the starting index may depend on a, H, B .

Proof. Direct differentiation gives

$$F'_n(x) = \frac{2(n-x)}{x} - \log \frac{ax^2}{n-x}, \quad (4.5)$$

$$F''_n(x) = -\frac{2n}{x^2} - \frac{2}{x} - \frac{1}{n-x} < 0. \quad (4.6)$$

Writing $r = (n-x)/x$, the equation $F'_n(x) = 0$ is precisely $r(1+r)e^{2r} = an$. Thus $x = j_n$ is the unique maximizer. At this point

$$F_n(j_n) = k_n(1 + 2r_n), \quad -F''_n(j_n) = \frac{\Delta_n}{k_n}. \quad (4.7)$$

Also $r_n \sim \frac{1}{2} \log n$, $j_n \sim 2n/\log n$, and $k_n \sim n$.

Stirling's formula, uniformly for j in a fixed multiplicative neighborhood of j_n , gives

$$\frac{(aj^2)^{n-j}}{(n-j)!} = \frac{e^{F_n(j)}}{\sqrt{2\pi(n-j)}} (1 + O(1/n)).$$

On $|j - j_n| \leq \sigma_n \log n$, the third derivative is $O(n/j_n^3)$, so the cubic Taylor error is $O((\log n)^3/\sqrt{n}) = o(1)$. The Gaussian lattice sum therefore gives

$$\sum_{|j-j_n| \leq \sigma_n \log n} \frac{(aj^2)^{n-j}}{(n-j)!} \sim \frac{e^{F_n(j_n)}}{\sqrt{2\pi k_n}} \sqrt{2\pi} \sigma_n = S_n.$$

The spacing divided by σ_n tends to zero. On the rest of a fixed multiplicative neighborhood, (4.6) bounds the drop in the exponent below by a constant times $(j - j_n)^2 (\log n)^2 / n$. The tail outside the displayed local window is therefore negligible even after multiplication by an arbitrary fixed power of n .

For completeness, we prove the stronger macroscopic assertion (4.4); it also controls the remaining bare tail. Choose a fixed $D \geq 1$ such that $aK^2 + BK + B \leq a(K + D)^2$ for all $K \geq 1$, put $x = K + D$ and $N = n + D - 1$, and let $h = \log H$. The logarithm of a summand is bounded above, up to $O(\log n)$, by

$$F_N(x) + hx,$$

where $N - x = n - K - 1$ is still an integer. The endpoint $N - x = 0$ has only exponential size and may be treated separately. This function is strictly concave. Its maximizer has $x \sim 2n/\log n$, because its critical equation is $r(1+r)e^{2r} = aNe^{-h}$. For each fixed $\alpha > 0$, elementary substitution gives

$$\frac{F_n(\alpha j_n) - F_n(j_n)}{n} \longrightarrow 2(\log \alpha - \alpha + 1). \quad (4.8)$$

At the two endpoints of I_n the limiting values correspond to $\alpha = 1/2$ and $\alpha = 2$ and are strictly negative. The added term hx is $o(n)$ there. Concavity places the maximum on either outside interval at its endpoint, for all large n . A fixed shift from n to N changes the maximum by only $O(\log n)$: differentiating the optimized exponent in n gives $2r_n = O(\log n)$. Thus every outside summand is at most $S_n e^{-c_1 n}$ up to a polynomial factor. Summing at most n terms proves (4.4) with a smaller positive c . $\square$

4.2 A coefficient lemma that retains signs

If $A(z) = \sum a_m z^m$ is analytic near zero, let $A^\#(t) = \sum |a_m| t^m$ inside a sufficiently small positive disc. This is only a majorant notation, not a new analytic operation outside its convergence disc.

Lemma 4.2 (Analytic-twist transfer). *Let Q, D be real analytic near zero, with*

$$Q(z) = z \exp(cz + O(z^2)), \quad D(z) = 1 + O(z).$$

Fix real d_0, e_0 and a finite lower cutoff J . Then

$$\sum_{j>J} [z^n] D(z) Q(z)^j e^{(aj^2 + d_0 j + e_0)z} \sim S_n \exp\left(\frac{c + d_0}{a} r_n\right). \quad (4.9)$$

Only $j \leq n$ contributes. In particular the left side is eventually positive, although the analytic factors can have signed coefficients.

Proof. Write $\log(Q(z)/z) = cz + G(z)$ with $G(z) = O(z^2)$, and let

$$k = n - j, \quad \Lambda_j = aj^2 + (c + d_0)j + e_0.$$

In the macroscopic window $j \asymp n/\log n$, $\Lambda_j > 0$ and $t = k/\Lambda_j = O((\log n)^2/n)$. The elementary coefficient identity

$$\frac{[z^k] A(z) e^{\Lambda_j z}}{\Lambda_j^k / k!} = \sum_{m=0}^k a_m \frac{(k)_m}{\Lambda_j^m} \quad (4.10)$$

shows that the absolute contribution of the nonconstant terms of $A(z) = D(z)e^{jG(z)}$ is at most

$$D^\#(t) e^{jG^\#(t)} - 1 = O(t + jt^2) = O((\log n)^3/n).$$

Here $(k)_m \leq k^m$; no positivity of A is assumed. Consequently, uniformly in the macroscopic window,

$$[z^n] D(z) Q(z)^j e^{(aj^2 + d_0 j + e_0)z} = \frac{(aj^2)^k}{k!} \exp\left(\frac{c + d_0}{a} \frac{k}{j}\right) (1 + o(1)). \quad (4.11)$$

The expansion of $k \log(\Lambda_j/(aj^2))$ has error $O(k/j^2) = O((\log n)^2/n)$.

On $|j - j_n| \leq \sigma_n \log n$,

$$\frac{k}{j} - r_n = O((\log n)^2 / \sqrt{n}) = o(1).$$

The exponential multiplier in (4.11) is therefore constant to relative $o(1)$ on the local saddle. Elsewhere in the macroscopic window it is bounded by a fixed power of n , so the Gaussian localization in Lemma 4.1 still applies.

We spell out why no uncontrolled contribution comes from smaller j . Choose a fixed disc where all analytic factors and their absolute coefficient majorants converge. For $j \leq C_0 \sqrt{n}$, a fixed-radius Cauchy bound gives $\exp(O(n))$, negligible compared with S_n . Increase C_0 so that for $j > C_0 \sqrt{n}$ the radius $t_0 = (n - j)/(aj^2)$ lies in that disc. Cauchy's estimate at t_0 bounds the absolute coefficient by

$$O(\sqrt{n+1}) \frac{(aj^2)^{n-j}}{(n-j)!} \exp\left(C_1 \frac{n-j}{j} + C_1\right).$$

For $j > C_0 \sqrt{n}$ the additional logarithm is $o(n)$; outside the macroscopic window it cannot defeat the exponential loss from Lemma 4.1. The case $j = n$ is a bounded endpoint. This proves (4.9) and eventual positivity. $\square$

Corollary 4.3 (Every fixed two-action-or-larger response has the right scale). *Under (2.1), for every fixed $M \geq 2$,*

$$A_{n,M} \sim L_n. \quad (4.12)$$

For $M = 1$ the corresponding equivalent is

$$A_{n,1} \sim S_n e^{(d-b)r_n/a}. \quad (4.13)$$

Proof. There are only finitely many exceptional tail slopes or weights; their individual terms are analytic and have coefficients of at most exponential size. Discard them at the large-order scale. Apply Lemma 4.2 with $c = -\mu$, $d_0 = d$ and $e_0 = e - b$. For $M = 1$, $Q_1 = ze^{-bz}$ and $D_1 = 1$, so $c = -b$. $\square$

5 A quantitative bound on multiple large actions

The response equivalent alone is not a proof of Theorem 2.1. This section supplies the missing estimate. Its explicit numerical constant 32 is deliberately conservative; it is not a sharp cutoff constant.

5.1 A positive majorant with a quadratic merging energy

Choose a fixed constant C such that

$$C \geq 2a + \max(d, 0), \quad |\lambda_{k+1} - b| \leq ak^2 + Ck \quad (k \geq 1). \quad (5.1)$$

Such a finite C exists under (2.1). For the pure model, $C = 2a$ works exactly. Define the positive formal model

$$P(z) = ze^{bz} + \sum_{k \geq 1} w_{k+1} P(z)^{k+1} e^{(ak^2 + Ck)z}. \quad (5.2)$$

For the same multiplicity vector as in (3.2), its coefficient contribution has positive energy

$$E_+(\mathbf{c}) = a \sum_k k^2 c_k + (C + b)K + b, \quad (5.3)$$

and equals

$$C(\mathbf{c})W(\mathbf{c}) \frac{E_+(\mathbf{c})^{n-K-1}}{(n-K-1)!}. \quad (5.4)$$

The same Lagrange calculation proves this expression. By (5.1), $|E_-(\mathbf{c})| \leq E_+(\mathbf{c})$. Thus it bounds the absolute value of each signed contribution, while preserving its number of large actions.

Let P_M be its finite core and T_M^+ its positive one-tail response. In particular

$$P_M(z) = z + \mu z^2 + O(z^3), \quad D_M^+(z) = 1 + O(z),$$

where the response denominator is now $1 - \sum_{j=2}^M j w_j P_M^{j-1} e^{(a(j-1)^2 + C(j-1))z}$. It is nonzero near zero and the corresponding Taylor coefficients are nonnegative. Put $A_{n,M}^+ = [z^n]T_M^+(z)$. The twist lemma gives

$$A_{n,M}^+ \sim S_n \exp\left(\frac{\mu + C - 2a}{a} r_n\right). \quad (5.5)$$

Lemma 5.1 (Total combinatorial weight at fixed excess). *There is a fixed $H \geq 1$ such that*

$$\sum_{\sum k c_k = K} C(\mathbf{c}) W(\mathbf{c}) \leq H^K \quad (K \geq 1). \quad (5.6)$$

One may use $H = 8 \max(1, \sup_j w_j)$.

Proof. The weights are bounded, and $s \leq K$. With weights removed, grouping ordered compositions by their number s gives

$$\sum_{\sum k c_k = K} C(\mathbf{c}) = \sum_{s=1}^K \frac{1}{K+1} \binom{K+s}{s} \binom{K-1}{s-1}.$$

Bound the first binomial by 4^K and sum the second to obtain 2^{K-1} . The result is at most 8^K . Restoring the weights contributes at most $\max(1, \sup_j w_j)^K$. $\square$

5.2 The merger estimate

Let $R_{n,M}^+$ be the sum of (5.4) over configurations with at least two excess indices $k \geq M$, counted with multiplicity.

Proposition 5.2 (Multiple-tail suppression). *For fixed $M \geq \max(2, J_0)$ with $M > 64$, there is $c > 0$ such that*

$$R_{n,M}^+ = O\left(n^{2-M/32} A_{n,M}^+\right) + O(S_n e^{-c n}). \quad (5.7)$$

All constants and starting indices may depend on the fixed model and on M .

Proof. We split the configurations into three sets.

1. Excess outside the saddle window. For every configuration,

$$E_+(\mathbf{c}) \leq aK^2 + (C+b)K + b.$$

Lemma 5.1 and (4.4) show that all $K \notin I_n$ contribute $O(S_n e^{-c_1 n})$.

2. Too much excess in bounded actions. Now let $K \in I_n$. Write K_c for the total excess in core actions $k < M$, and let $L = K - K_c$ be the tail excess. If $K_c > K/4$, then $L < 3K/4$ and

$$\sum_k k^2 c_k \leq L^2 + (M-1)K_c \leq \frac{9}{16} K^2 + (M-1)K.$$

For all sufficiently large n , this gives $E_+(\mathbf{c}) \leq (5/8)aK^2$. At fixed K , use (5.6) and compare with the single bare term of action $K+1$. Writing $N = n - K - 1$, the ratio is at most $H^K (5/8)^N$. Uniformly on I_n , $K = O(n/\log n)$ and $N \sim n$, so this ratio is exponentially small in n . Summing over K contributes another $O(S_n e^{-c_2 n})$.

3. One merged tail carrying most of the excess. It remains to treat $K \in I_n$ and $L \geq 3K/4$. Fix the core multiplicities. Suppose the tail has $r \geq 2$ ordered excesses $k_1, \dots, k_r \geq M$ with total L . Merge them into the single excess L , leaving all core actions unchanged. This is a valid one-tail configuration. Its energy E' satisfies

$$E' - E_+ = a \left(L^2 - \sum_{i=1}^r k_i^2 \right),$$

$$L^2 - \sum_{i=1}^r k_i^2 \geq (r-1)M(2L - rM) \geq (r-1)ML. \quad (5.8)$$

For the first inequality, at fixed sum and lower bound the sum of squares is largest when $r-1$ entries equal M ; transferring mass from a smaller to a larger entry proves this extremal assertion. The second uses $rM \leq L$.

For large n , $E' \leq 2aK^2$ and $N \geq n/2$. Hence

$$\left(\frac{E_+}{E'} \right)^N \leq \exp \left(-\frac{Na(r-1)ML}{E'} \right)$$

$$\leq \exp \left(-\frac{3M(r-1)n}{16K} \right) \leq n^{-M(r-1)/32}. \quad (5.9)$$

Indeed $K \leq 4n/\log n$ gives the stronger exponent $3M(r-1)/64$; we have weakened it to $M(r-1)/32$ for convenience.

Let s_c be the number of core actions. The ratio of the factorial combinatorial factors before and after merging is

$$\frac{(K + s_c + r)!}{(K + s_c + 1)! \prod_{k \geq M} c_k!} \leq \frac{(2K)^{r-1}}{\prod_{k \geq M} c_k!}.$$

The tail weights are all one because $M \geq J_0$. For fixed L, r ,

$$\sum_{\substack{\sum k c_k = L \\ \sum c_k = r, k \geq M}} \frac{1}{\prod c_k!} = \frac{1}{r!} \# \{ (k_1, \dots, k_r) : k_i \geq M, \sum k_i = L \} \leq \frac{L^{r-1}}{r!}.$$

Combining these bounds, the total contribution mapping to a particular merged configuration is at most its one-tail contribution multiplied by

$$\sum_{r \geq 2} \frac{(2n^{2-M/32})^{r-1}}{r!} = O(n^{2-M/32}),$$

since $M > 64$. Sum over all possible merged configurations. This proves (5.7) together with the first two parts. $\square$

5.3 From a coarse large core to the signed inverse

Set

$$\nu = \frac{2\mu + C - 2a - d}{2a} \geq 0. \quad (5.10)$$

The nonnegativity follows from the choice of C .

Theorem 5.3 (Arbitrary algebraic accuracy from a sufficiently large fixed core). *For fixed $M \geq \max(2, J_0)$ with $M > 64$,*

$$|v_n - [z^n](Q_M - T_M)| = O(L_n n^{2-M/32+\nu}) + O(S_n e^{-cn}). \quad (5.11)$$

In particular, for each fixed $A > 0$, a fixed core satisfying $M > 32(A + 2 + \nu)$ and $M \geq J_0$ makes the difference $o(L_n n^{-A})$.

Proof. By the marked identity, the difference contains exactly the configurations with at least two large actions. Its absolute value is bounded by $R_{n,M}^+$. Apply Proposition 5.2 and (5.5). The ratio of the latter scale to L_n is

$$\exp\left(\frac{2\mu + C - 2a - d}{a}r_n\right) = e^{2\nu r_n} = \left(\frac{an}{r_n(1+r_n)}\right)^\nu = O(n^\nu).$$

This proves (5.11). Every fixed power of n times S_n/L_n is subexponential in n , so the last term is negligible at the stated relative scales. $\square$

Proof of Theorem 2.1. Choose a fixed M large enough that $2 - M/32 + \nu < 0$. Theorem 5.3 makes all multiple-tail terms $o(L_n)$. Since Q_M is analytic, $[z^n]Q_M = O(\rho^{-n})$ for some $\rho > 0$, which is also $o(L_n)$. Corollary 4.3 now gives $v_n \sim -A_{n,M} \sim -L_n$.

To obtain the ratio assertion, extend r_n, j_n, k_n, S_n, L_n smoothly to positive real n . Implicit differentiation yields

$$r'_n = \frac{1}{n(2 + 1/r_n + 1/(1+r_n))}. \quad (5.12)$$

The derivative of the optimized exponent $F_n(j_n)$ with respect to n is $2r_n$. Thus

$$\frac{d}{dn} \log L_n = 2r_n + O(1/n).$$

Integrating over a unit interval gives $L_n/L_{n-1} \sim e^{2r_n}$. The equivalent for v_n transfers this ratio and proves (2.7). $\square$

6 Sharp small-core errors and finite-perturbation universality

6.1 Comparing analytic cores before taking coefficients

The large cutoff in Theorem 5.3 is a proof device. It does not describe the smallest useful approximation. We now recover the precise error of any fixed core containing the second action.

Lemma 6.1 (First difference of two cores). *Fix $M \geq 2$, let $\ell = \min\{j > M : w_j > 0\}$, and fix $N \geq \ell$. Then, as convergent Taylor expansions near zero,*

$$Q_N - Q_M = -w_\ell z^\ell + O(z^{\ell+1}), \quad (6.1)$$

$$\log(Q_N/Q_M) = -w_\ell z^{\ell-1} + O(z^\ell), \quad (6.2)$$

$$D_N/D_M = 1 + O(z^{\ell-1}). \quad (6.3)$$

The quotient Q_N/Q_M is understood after canceling its common simple zero at the origin.

Proof. The equations for the two cores agree through degree $\ell - 1$. Their first extra summand is $-w_\ell Q_N^\ell e^{(\lambda_\ell - b)z} = -w_\ell z^\ell + O(z^{\ell+1})$. The derivative with respect to the unknown in the common implicit equation has constant coefficient one. This proves (6.1). Dividing by $Q_M = z + O(z^2)$ and taking the analytic logarithm proves (6.2). In the response denominator, the first new primitive term has order $\ell - 1$. Substitution of $Q_N - Q_M$ in the old denominator changes it only at order at least ℓ . This proves (6.3). $\square$

Lemma 6.2 (Transfer of a core difference). *With the notation of Lemma 6.1,*

$$A_{n,N} - A_{n,M} \sim -w_\ell j_n t_n^{\ell-1} L_n. \quad (6.4)$$

Proof. Write $s = \ell - 1 \geq 2$. The finitely many summands with $M < j \leq N$ are analytic functions, so their coefficients have at most exponential size. The same holds for the finitely many exceptional slopes in the common tail. They are negligible at the scale on the right of (6.4). We may therefore compare the summands for j larger than all these fixed indices.

Set

$$\log(Q_M/z) = -\mu z + G(z), \quad G(z) = O(z^2), \quad \Lambda_j = aj^2 + (d - \mu)j + e - b.$$

After extracting $z^j e^{\Lambda_j z}$, the difference of the remaining analytic factors is

$$D_M(z) e^{jG(z)} \left\{ \frac{D_N(z)}{D_M(z)} \exp\left(j \log \frac{Q_N(z)}{Q_M(z)}\right) - 1 \right\}. \quad (6.5)$$

By Lemma 6.1, the expression in braces is $-w_\ell z^s + E_j(z)$. In a sufficiently small disc, its absolute coefficient majorant at a positive radius t satisfies

$$E_j^\#(t) \leq C(t^s + jt^{s+1} + j^2 t^{2s}) \exp(Cjt^s), \quad (6.6)$$

where C depends only on the two fixed cores. This follows by expanding $D_N/D_M = 1 + O(z^s)$ and $\log(Q_N/Q_M) = -w_\ell z^s + O(z^{s+1})$, and bounding the quadratic and higher terms of the exponential by their absolute Taylor series.

In the macroscopic saddle window, put $k = n - j$ and $t = k/\Lambda_j$. Then $t \asymp (\log n)^2/n$, $jt^2 = O((\log n)^3/n) = o(1)$, and $jt^s = o(1)$. Apply (4.10) to (6.5). For the leading term, the shift by s gives

$$\frac{[z^k] D_M(z) e^{jG(z)} z^s e^{\Lambda_j z}}{\Lambda_j^k / k!} = \frac{(k)_s}{\Lambda_j^s} (1 + O(t + jt^2)) = t^s (1 + O(1/k + t + jt^2)).$$

The same identity and (6.6) bound the error, relative to $jt^s \Lambda_j^k / k!$, by

$$O(1/j + t + jt^s) \exp(O(jt^s)) D_M^\#(t) e^{jG^\#(t)} = o(1).$$

Crucially, the common analytic factor is applied to the *difference* which already vanishes to order s . We have not subtracted two unrelated relative $o(1)$ estimates. Such a subtraction would not establish (6.4) when s is large.

Thus the common-tail coefficient difference, uniformly in the saddle window, is

$$-w_\ell j \left(\frac{k}{\Lambda_j} \right)^s \frac{\Lambda_j^k}{k!} (1 + o(1)). \quad (6.7)$$

On $|j - j_n| \leq \sigma_n \log n$, the multiplier $j(k/\Lambda_j)^s$ is $j_n t_n^s (1 + o(1))$. The remaining exponential and factorial factor is handled by Lemma 4.2. Within the rest of the macroscopic window, the absolute versions of the preceding bounds are a fixed polynomial multiple of the bare summand. Its Gaussian tail is smaller than every fixed power of n . Outside the macroscopic window, the Cauchy estimates from the proof of Lemma 4.2, applied separately to the two cores, give $O(S_n e^{-cn+o(n)})$. Both tails are negligible even relative to $L_n j_n t_n^s$, since s is fixed. Summing (6.7) proves (6.4). $\square$

Proof of Theorem 2.2. Fix M and ℓ as in the statement. Choose a fixed auxiliary core $N \geq \max(J_0, \ell)$ large enough that Theorem 5.3 yields

$$v_n = -A_{n,N} + o(L_n j_n t_n^{\ell-1}).$$

This is possible because $j_n t_n^{\ell-1}$ is a fixed power of n times a power of $\log n$, and because the analytic coefficient of Q_N is negligible. By Lemma 6.2,

$$A_{n,N} = A_{n,M} - w_\ell j_n t_n^{\ell-1} L_n + o(L_n j_n t_n^{\ell-1}).$$

Finally, $A_{n,M} \sim L_n > 0$ by Corollary 4.3. Substitution proves (2.11), including its sign. In the pure case, direct substitution from (2.4) gives (2.12). $\square$

6.2 A minimal explicit two-action core

For the pure model the smallest core with the correct leading equivalent has a particularly simple closed form:

$$Q_2(z) = \frac{2ze^{-az}}{1 + \sqrt{1 + 4ze^{2az}}}, \quad (6.8)$$

$$D_2(z) = (1 + 4ze^{2az})^{-1/2}, \quad (6.9)$$

$$T_2(z) = D_2(z) \sum_{j \geq 3} Q_2(z)^j e^{a(j^2-1)z}. \quad (6.10)$$

All square roots are the analytic branches with value one at zero. Indeed, $Q_2 + e^{3az}Q_2^2 = ze^{-az}$ is a quadratic equation, and $1 + 2e^{3az}Q_2 = \sqrt{1 + 4ze^{2az}}$.

Corollary 6.3 (Explicit minimal-core accuracy). *For the pure quadratic model,*

$$v_n = -[z^n]T_2(z) \left(1 - \frac{r_n^2(1 + r_n)}{a^2n} + o\left(\frac{r_n^2(1 + r_n)}{n}\right) \right). \quad (6.11)$$

The relative defect is asymptotic to $(\log n)^3/(8a^2n)$. In contrast, the one-action response is not a relative approximation:

$$\frac{A_{n,1}}{-v_n} \sim e^{r_n/a} \rightarrow \infty. \quad (6.12)$$

Proof. The two-action assertion is Theorem 2.2 with $M = 2$. The one-action assertion follows from (4.13) and Theorem 2.1, since $b = a$ and $w_2 = 1$. $\square$

The response in (6.10) is still a formal series. The elementary closed form of its core does not make the infinite positive-argument tail convergent. Coefficients through degree n require only $j \leq n$.

6.3 Which finite changes matter

Corollary 6.4 (Finite-perturbation universality). *Consider two families satisfying (2.1), with the same a, d and the same value of $\mu = \lambda_1 + w_2$. Their inverse coefficients satisfy*

$$\frac{v_n^{(1)}}{v_n^{(2)}} \rightarrow 1. \quad (6.13)$$

Their eventual constants e , all finitely many slopes of index at least two, and all finitely many weights of index at least three may differ. For the same a, d but different μ , the ratio is instead

$$\frac{v_n^{(1)}}{v_n^{(2)}} \sim \exp\left(-\frac{\mu_1 - \mu_2}{a} r_n\right). \quad (6.14)$$

Proof. Divide the two instances of (2.6). $\square$

When $w_2 = 0$, the one-action core already has the correct quadratic jet, and (4.13) agrees with the full equivalent. Thus the minimal-core statement above is a statement about the pure family, not an assertion that a nonzero second action must exist in every perturbation. If the first omitted nonzero action has index at least three, the proof of Theorem 2.2 also works for $M = 1$. The condition that matters in that proof is $s = \ell - 1 \geq 2$ together with the common quadratic jet.

7 A borderline tail perturbation changes the leading constant

The assumption that the tail is exactly affine-quadratic is stronger than needed, but weakening it merely to $\lambda_j \sim aj^2$ would lose multiplicative information. The next theorem identifies a sharp, simple intermediate scale.

Theorem 7.1 (A logarithmic boundary for slope perturbations). *Keep $w_1 = 1$, $w_j, \lambda_j \geq 0$, and $w_j = 1$ eventually. Suppose that, for fixed $a > 0, d, e, \kappa \in \mathbb{R}$,*

$$\lambda_j = aj^2 + dj + e + \varepsilon_j, \quad \frac{\varepsilon_j \log j}{j} \longrightarrow \kappa. \quad (7.1)$$

There is no monotonicity or differentiability assumption on ε_j . With $\mu = \lambda_1 + w_2$ and the same r_n, S_n ,

$$v_n \sim -e^{\kappa/(2a)} S_n \exp\left(\frac{d - \mu}{a} r_n\right). \quad (7.2)$$

The sharp first-omitted-action law remains valid, with $A_{n,M}$ formed from the actual perturbed kernel. The inverse Borel equivalent (2.15) acquires the same multiplicative constant $e^{\kappa/(2a)}$.

Proof. Since $\varepsilon_j = o(j)$, a finite constant C satisfying (5.1) still exists. The entire multiple-tail majorant and merger proof remain valid without change. It remains to check the one-tail equivalent; neither that argument nor the comparison of cores requires a derivative of the tail slopes.

In the macroscopic saddle window, replace Λ_j in the twist proof by $\Lambda_j + \varepsilon_j$. Its ratio to aj^2 still tends uniformly to one. The extra logarithm in the exponential coefficient is

$$k \log \left(1 + \frac{\varepsilon_j}{\Lambda_j}\right) = \frac{k\varepsilon_j}{aj^2} + o(1). \quad (7.3)$$

Here the quadratic error and the error in replacing Λ_j by aj^2 are $o(1)$ uniformly. The hypothesis in (7.1) means a uniform tail bound over all sufficiently large integer j ; it is therefore uniform on the moving saddle window as well. On $|j - j_n| \leq \sigma_n \log n$,

$$\frac{k\varepsilon_j}{aj^2} = \frac{\kappa + o(1)}{a} \frac{k}{j \log j} \longrightarrow \frac{\kappa}{2a}.$$

Indeed $k/j = r_n + o(1)$ and $\log j_n \sim 2r_n$. On the whole macroscopic window this extra exponent is bounded. Outside that window, $\varepsilon_j = o(j)$ is absorbed by the Cauchy bounds already used. Thus the twist equivalent is multiplied by $e^{\kappa/(2a)}$.

The coarse signed reduction proves (7.2). For the sharp defect, compare two cores as before; their tail is common, so the additional factor appears in both the leading response and its difference and cancels in their ratio. The Borel statement follows from the transfer theorem proved in the next section, since a fixed coefficient multiplier is preserved by that theorem. $\square$

Example 7.2 (Asymptotically equal slopes need not give equivalent coefficients). Start with the pure slopes, retain any finite initial set unchanged, and for all sufficiently large j replace aj^2 by $aj^2 + \kappa j / \log j$. These slopes remain nonnegative after enlarging the retained initial set, even for negative κ . They are asymptotic to aj^2 , have the same $a, d = 0$ and the same finite-core parameter μ , but their inverse coefficients have ratio tending to $e^{\kappa/(2a)}$ against the pure model. Thus leading quadratic growth of the slopes alone does not determine the leading inverse constant.

8 Borel growth from a second moving saddle

8.1 A transfer theorem for the moving factorial scale

The coefficient asymptotic is strong enough to determine the positive-ray Borel transform to relative error tending to zero. The following formulation also permits a useful comparison with the forward coefficients.

Theorem 8.1 (Sharp Borel transfer). *Fix real constants A, B and suppose that a real sequence c_n is eventually positive and satisfies*

$$c_n \sim S_n \exp(Ar_n^2 + Br_n). \quad (8.1)$$

Then $\sum_{n \geq 1} c_n t^n / n!$ is entire. With R as in (2.14),

$$\sum_{n \geq 1} \frac{c_n t^n}{n!} \sim \frac{\exp\left(Re^{2R}/a + AR^2 + BR\right)}{\sqrt{2R+1}} \quad (t \rightarrow +\infty). \quad (8.2)$$

The finitely many initial terms may have arbitrary signs.

Proof. We first prove convergence. The saddle identity gives

$$\begin{aligned} \frac{1}{n} \log S_n &= \frac{r_n}{1+r_n} (1+2r_n) + o(1) \\ &= 2r_n - 1 + \frac{1}{1+r_n} + o(1). \end{aligned} \quad (8.3)$$

Since $r_n = O(\log n)$, the extra terms $Ar_n^2 + Br_n$ are $o(n)$. Stirling's formula and $e^{2r_n} = an/[r_n(1+r_n)]$ imply

$$\left(\frac{|c_n|}{n!}\right)^{1/n} \sim \frac{a}{r_n(1+r_n)} \sim \frac{4a}{(\log n)^2} \rightarrow 0. \quad (8.4)$$

This proves entire convergence.

It suffices next to analyze the sequence exactly equal to the right side of (8.1). Extend the saddle parameters smoothly to real $x > 0$ and put

$$H_t(x) = F_x(j_x) + x(1 + \log(t/x)). \quad (8.5)$$

The optimized first phase satisfies $\partial_x F_x(j_x) = 2r_x$, and therefore

$$H'_t(x) = 2r_x + \log(t/x) = \log \frac{at}{r_x(1+r_x)}, \quad (8.6)$$

$$H''_t(x) = -\frac{2r_x+1}{x\Delta_x} < 0. \quad (8.7)$$

The last equality follows directly from (5.12). The unique maximum occurs at

$$r_{x_*} = R, \quad x_* = te^{2R} = \frac{R(1+R)e^{2R}}{a}. \quad (8.8)$$

At that point,

$$H_t(x_*) = \frac{Re^{2R}}{a}, \quad \tau_*^2 := \frac{1}{-H''_t(x_*)} = \frac{x_*\Delta(R)}{2R+1}. \quad (8.9)$$

In particular $\tau_* \asymp \sqrt{x_* R} \rightarrow \infty$ and $\tau_*/x_* \rightarrow 0$.

Stirling's formula gives the model summand, uniformly near x_* , as

$$\frac{\exp(H_t(n) + Ar_n^2 + Br_n)}{\sqrt{2\pi n \Delta_n}} (1 + O(1/n)). \quad (8.10)$$

On $|n - x_*| \leq \tau_* \log x_*$,

$$r_n - R = O\left(\sqrt{R/x_*} \log x_*\right) = o(1/R).$$

Consequently $Ar_n^2 + Br_n = AR^2 + BR + o(1)$ there, and the algebraic amplitude is constant to relative $o(1)$. Differentiating (8.7) shows that $H_t'''(x) = O(1/(x_*^2 R))$ uniformly on $[x_*/2, 2x_*]$. The third-order Taylor error on the local window is therefore $O(\sqrt{R/x_*}(\log x_*)^3) = o(1)$. The local Gaussian lattice sum is

$$\begin{aligned} & \frac{\exp(H_t(x_*) + AR^2 + BR)}{\sqrt{2\pi x_* \Delta(R)}} \sqrt{2\pi} \tau_* \\ &= \frac{\exp(Re^{2R}/a + AR^2 + BR)}{\sqrt{2R + 1}}. \end{aligned} \quad (8.11)$$

We justify localization also for the infinite right tail. On $[x_*/2, 2x_*]$, the negative curvature is bounded above and below by positive multiples of $1/(x_* R)$, so the complement of the local window has an exponential Gaussian loss. The extra logarithm $Ar_x^2 + Br_x$ has derivative $O(R/x_*)$ in this interval, which cannot offset that loss. At $x = x_*/2$ and $x = 2x_*$, the phase drop is at least cx_*/R . Beyond $2x_*$, strict concavity and $H_t'(2x_*) \leq -c_1/R$ bound the tail by a geometric series with ratio at most $e^{-c_1/(2R)}$, after increasing t if needed. To see that the extra logarithm does not invalidate this last bound, note that

$$\left| \frac{d}{dx} (Ar_x^2 + Br_x) \right| = O((1 + r_x)/x).$$

For all $x \geq 2x_*$ its supremum tends to zero much faster than $1/R$. The variation of the algebraic amplitude is smaller still. The left tail below $x_*/2$ is bounded similarly by concavity; the extra logarithm is $O((\log x_*)^2)$ there. A fixed finite set of small indices contributes only a polynomial in t . These estimates prove that the tails are negligible relative to (8.11).

Finally, a coefficient equivalent transfers through the sum without any rate assumption. Given $\epsilon > 0$, all sufficiently large n have c_n between $1 - \epsilon$ and $1 + \epsilon$ times the positive model coefficient. The omitted finite set contributes only a polynomial in t , whereas (8.11) grows faster than every polynomial. Squeezing and then letting $\epsilon \downarrow 0$ proves (8.2). $\square$

8.2 Application to the inverse and the obstruction to positive summation

Proof of Theorem 2.3. Theorem 2.1 makes $c_n = -v_n$ eventually positive, with $A = 0$ and $B = (d - \mu)/a$ in Theorem 8.1. This proves both entire convergence and (2.15).

Let $M_V(t) = \max_{|z|=t} |\mathcal{B}_V(z)|$. The positive-ray lower bound gives

$$\log \log M_V(t) \geq 2R + O(\log R), \quad R \sim \sqrt{at}.$$

Hence the entire-function order $\limsup_{t \rightarrow \infty} \log \log M_V(t) / \log t$ is infinite. Furthermore, $\mathcal{B}_V(t) < 0$ for all sufficiently large t , and its absolute value grows faster than e^{ct} for every fixed $c > 0$. For any real $z > 0$, the integral

$$\frac{1}{z} \int_0^\infty e^{-t/z} \mathcal{B}_V(t) dt \quad (8.12)$$

diverges to $-\infty$. This is failure of the ordinary positive-direction Borel–Laplace prescription in our normalization, not a claim that no summation method or other direction can work. $\square$

8.3 A forward–inverse Borel separation law

The following corollary is the only result in this paper that additionally uses a nontrivial asymptotic theorem from the earlier repository manuscript. It can also be read as a conditional transfer statement given the displayed forward coefficient equivalent.

Corollary 8.2 (Exponential separation of the two positive Borel transforms). *For the pure model, use the forward coefficient theorem of [2], namely*

$$u_n \sim S_n \exp\left(\frac{a+1}{a} r_n^2\right). \quad (8.13)$$

Let $\mathcal{B}_U(t) = \sum_{n \geq 1} u_n t^n / n!$. Then

$$\mathcal{B}_U(t) \sim \frac{\exp\left(Re^{2R}/a + (a+1)R^2/a\right)}{\sqrt{2R+1}}, \quad (8.14)$$

$$\frac{-\mathcal{B}_V(t)}{\mathcal{B}_U(t)} \sim e^{-(a+1)t}. \quad (8.15)$$

Proof. Apply Theorem 8.1 to (8.13) with $A = (a+1)/a$ and $B = 0$. Divide the inverse equivalent by the resulting forward equivalent. The quotient of the exponential factors is

$$\exp\left(-\frac{a+1}{a}(R^2 + R)\right) = e^{-(a+1)t},$$

using $R(1+R) = at$. □

Both Borel transforms have the same dominant double-exponential growth, but their ratio retains the exact exponential penalty (8.15). Thus agreement at the logarithm-of-logarithm level would conceal a substantial effect of reversion.

9 Exact algorithms and completed computational checks

9.1 An integer-normalized triangular recurrence

For integer slopes and weights, the inverse can be computed entirely with integers. Set

$$c_n = n!v_n, \quad P_{j,m} = m![z^m]\widehat{V}(z)^j, \quad P_{0,0} = 1, \quad P_{0,m} = 0 \ (m > 0).$$

Then $c_1 = 1$, $P_{1,m} = c_m$, and the normalized inverse equation gives

$$P_{j,m} = \sum_{i=1}^{m-j+1} \binom{m}{i} c_i P_{j-1,m-i} \quad (j \geq 2), \quad (9.1)$$

$$c_n = n(-b)^{n-1} - \sum_{j=2}^n w_j \sum_{m=j}^n \binom{n}{m} P_{j,m} (\lambda_j - b)^{n-m}. \quad (9.2)$$

Only already known c_i , $i < n$, occur on the right when computing c_n . Thus negative exponent coefficients $\lambda_j - b$, when present, cause no algebraic difficulty or loss of exactness.

The program also evaluates this recurrence over the dual-number ring $\mathbb{Z}[h]/(h^2)$, multiplying every primitive weight with $j > M$ by h . Its constant component is $n![z^n]Q_M$ and its linear component is $-n!A_{n,M}$. This computes the one-tail response without differentiating an implicitly defined analytic function numerically.

The direct dynamic program uses $O(N^3)$ arithmetic operations and $O(N^2)$ stored integers through order N for each fixed core. This is an arithmetic-operation count, not a bit-complexity estimate; integer sizes grow rapidly. A production implementation could exploit faster polynomial multiplication, but the transparent recurrence is preferable for the independent checks supplied here.

9.2 Independent finite checks and their precise scope

Two completed runs are included. For the pure models $a = 1$ and $a = 2$, the program computed the full inverse and the $M = 2, 3$ marked cores through orders 240 and 180, respectively. In each run it also computed a finite perturbation through order 120:

$$\lambda_2 = 4a + 1, \quad w_3 = 2,$$

with all other pure parameters unchanged. This leaves $\mu = \lambda_1 + w_2 = a + 1$ fixed.

Each run passed exactly 134 finite assertions, or 268 assertions over the two runs. Their decomposition is as follows:

Independent comparison in one run	Assertions
The multiplicity formula against the inverse recurrence, degrees 1–16	16
The multiplicity formula against each of the constant and linear marked components, two cores, degrees 1–16	64
Ordinary rational-series substitution in $\sum_j w_j V^j e^{\lambda_j z} = z$, degrees 0–20	21
The multiplicity formula for the modified model, degrees 1–16	16
Ordinary rational-series substitution for the modified model, degrees 0–16	17
Total	134

The independent substitution uses ordinary `Fraction` arithmetic, not the integer-normalized inverse recurrence. The partition enumeration uses the finite formula (3.5) rather than the recurrence. The high-order coefficients beyond those small independent check ranges were computed by the recurrence; they are not claimed independently certified through the full recorded order.

For example, the pure $a = 1$ inverse begins

$$\hat{V}(z) = z - 2z^2 + \frac{1}{2}z^3 - \frac{2}{3}z^4 - \frac{29}{8}z^5 - \frac{743}{60}z^6 - \frac{8491}{144}z^7 + O(z^8). \quad (9.3)$$

In particular the coefficient of z^3 is positive. The theorem proves *eventual* negativity, not negativity of every coefficient after the linear term. The recorded audit files deliberately report that the latter stronger finite assertion is false.

9.3 Diagnostics for the equivalent and the first omitted action

Define the pure-model diagnostic ratios

$$E_n = \frac{v_n}{-L_n}, \quad C_{n,M} = \frac{-v_n}{A_{n,M}}, \quad D_{n,M} = \frac{1 - C_{n,M}}{j_n t_n^M}.$$

Theorems 2.1 and 2.2 predict that each tends to one. The following entries use exact integer coefficients with 70-digit `mpmath` evaluation of the transcendental saddle expressions. The displayed decimals are rounded diagnostics, not interval enclosures.

The elementary leading equivalent converges noticeably more slowly than the exact two-action response. The three-action response is more accurate still: for $a = 1, n = 240$, its ratio $C_{240,3}$ is approximately 0.99873715, whereas $C_{240,2}$ is approximately 0.95257828. The normalized defects are also approaching one, though the third-core normalization has substantial finite-order corrections at the recorded orders. These data support the normalization and sign in the sharp law; no finite table proves an asymptotic limit.

Table 1: Completed exact-coefficient runs and asymptotic diagnostics. All four diagnostic columns have predicted limit one.

a	n	E_n	$C_{n,2}$	$D_{n,2}$	$D_{n,3}$
1	40	0.6077450	0.8550193	1.4978435	2.5942016
1	80	0.7259650	0.9081279	1.2630352	1.7680004
1	120	0.7804187	0.9283502	1.1840540	1.5260765
1	240	0.8525778	0.9525783	1.1009614	1.2871477
2	40	0.6567062	0.9300627	1.9229610	3.8557558
2	80	0.7682482	0.9610937	1.4752948	2.2258330
2	120	0.8175957	0.9713215	1.3316234	1.8064854
2	180	0.8574533	0.9785043	1.2340344	1.5472156

The finite-perturbation diagnostic also points in the predicted direction. For $a = 1$, the modified-to-pure inverse ratios at $n = 40, 80, 120$ are approximately 0.790293, 0.847742, and 0.875324. For $a = 2$ they are 0.897186, 0.934662, and 0.949539. Their common predicted limit is one by Corollary 6.4. No monotonic convergence theorem is asserted from these observations.

9.4 Reproducibility and what was not computed

The companion archive contains `verification/verify.py`, the two sets of exact coefficients, their decimal diagnostic tables, and the machine-readable assertion reports. From the archive root the recorded runs can be reproduced with

```
python verification/verify.py --a 1 --order 240 --cores 2 3
--out verification/results_a1

python verification/verify.py --a 2 --order 180 --cores 2 3
--out verification/results_a2
```

Each displayed command is a single logical command; the line break above is typographical. Python 3.10 or newer is required for the type syntax. The exact routines use only the standard library. The command-line audit also requires `mpmath` for the decimal diagnostics.

No numerical positive-ray Borel growth experiment is used to establish Theorem 2.3: it follows from the coefficient theorem and the second-saddle proof. No interval-certified finite starting index, automated proof-assistant check, or numerical verification at unreported higher orders is claimed. Exact finite arithmetic tests the formulas and implementation, not the infinite asymptotic argument.

10 Further research questions and concrete next targets

10.1 Subquadratic feedback and a different small-core hierarchy

For slopes $\lambda_j = aj^p$, $1 < p < 2$, the bare saddle scales suggest $j_n \asymp n/\log n$ and $t_n \asymp n^{1-p}(\log n)^p$. A degree- m term in the logarithm of an inverse core is multiplied by $j_n t_n^m$, whose scale is

$$j_n t_n^m \asymp n^{1-m(p-1)}(\log n)^{pm-1}. \quad (10.1)$$

This calculation suggests an inverse-core hierarchy different from the forward hierarchy, and logarithmic boundary effects when $m(p-1) = 1$. It is not a theorem extending the present paper: the cancellation penalty can now be larger than a fixed power of n , while our coarse merger estimate is polynomial for a fixed cutoff. A worthwhile target is a signed finite-core reduction

that is exponentially accurate on the scale of that penalty, followed by the appropriate moving saddle.

10.2 Growing cores and effective error certificates

Every cutoff in this article is fixed as $n \rightarrow \infty$. Can one allow $M = M_n \rightarrow \infty$ while retaining uniform analytic-core radii and explicit constants? Such a result would turn arbitrary algebraic accuracy into an algorithm with a chosen tolerance. It would need bounds on the dependence of the implicit-function disc, the Taylor majorants, and the merger starting index on M . The conservative constant 32 in Theorem 5.3 is also a natural optimization target, although improving it alone would not provide uniformity.

A particularly useful intermediate objective is a verified finite- n enclosure for $v_n + A_{n,2}$ with the correct sign and a computable relative bound around $A_{n,2}j_n t_n^2$. The current proof gives eventual sign and an asymptotic constant, but not a numerical universal threshold.

10.3 All-order relative expansions within a fixed core

Theorem 2.2 isolates the first omitted action. Higher terms should combine the next omitted actions, repeated copies of the first one, response-denominator corrections, and higher derivatives of the saddle. These sources have different powers of n and $\log n$ and can resonate. A constructive goal is an explicit coefficient algebra that orders these contributions and outputs every term above a prescribed scale.

Theorem 5.3 already proves that a sufficiently large fixed core captures any prescribed algebraic accuracy. What remains is to turn the finite-core coefficient extraction itself into a complete relative expansion with a specified remainder, rather than only the leading term or first core difference.

10.4 Beyond the logarithmic slope boundary

Theorem 7.1 handles $\varepsilon_j \log j/j \rightarrow \kappa$ without regularity assumptions. For larger but still $o(j)$ perturbations, the natural candidate factor is

$$\exp\left(\frac{r_n \varepsilon_{j_n}}{a j_n}\right),$$

provided the perturbation varies sufficiently little over the moving action window. This expression is only a proposed diagnostic when noninteger j_n requires interpolation. One should formulate the correct discrete modulus-of-variation hypothesis and determine when a saddle displacement also contributes at multiplicative order.

An oscillatory perturbation on the Gaussian window could instead produce an averaged profile or destroy a single leading constant. Explicit necessary and sufficient conditions for these alternatives would extend the universality statement beyond polynomial tails.

10.5 Complex Borel growth and admissible summation directions

The positive-ray Borel growth is now sharp. The next question is the complex-direction geometry: how do the square-root branch in $R(t)$, the second saddle, and the eventual signed coefficients combine away from the positive ray? Determine sectors of controlled growth, possible Stokes changes between competing saddles, and the asymptotic zero distribution of $\mathcal{B}_V$.

Entire Borel continuation and failure of the positive Laplace integral do not determine any of these questions. In particular this article proves neither a global complex growth formula nor a

new directional summability theorem. A first tractable step is a uniform complex saddle theorem in a shrinking angular neighborhood of the positive ray, where the real saddle can be continued without encountering other stationary points.

10.6 Signed and complex primitive parameters

Nonnegative weights made the positive merger comparison transparent. With finitely many signed or complex primitive weights, the analytic core still exists, but its leading factor can have a phase and the comparison scale may change. It is natural to ask whether the same finite-core principle survives under a quantitative noncancellation condition. Allowing signs throughout the infinite tail is a more substantial change: a large-action cancellation can alter the dominant saddle or remove its leading contribution completely.

10.7 A staged formalization program

The exact portion is a reasonable first formalization target: the triangular inverse equation, the multiplicity identity, the marked coefficient decomposition, and the finite inequality

$$L^2 - \sum_{i=1}^r k_i^2 \geq (r-1)M(2L - rM) \quad (k_i \geq M, \sum_i k_i = L).$$

These are independent of delicate asymptotic analysis. The next layer is uniform coefficient majorization and the discrete Gaussian sum, followed by the two saddle transfer lemmas. Only after those layers should the signed finite-core theorem and the Borel conclusion be assembled.

No Lean implementation is included here. Existing repository asymptotic or well-based-support results may supply useful infrastructure, but their presence does not establish any of the new inequalities or transfer lemmas automatically.

11 Proof dependencies, novelty boundaries, and conclusion

The logical route to the main inverse theorem is

$$\begin{aligned} \text{inverse kernel} &\longrightarrow \text{extexactmarkedtreeexpansion} \\ &\longrightarrow \text{extpositivemergerbound} \\ &\longrightarrow \text{extsignedone} - \text{tailequivalence}. \end{aligned}$$

The analytic-twist lemma supplies the one-tail coefficient scale along this route. The sharp defect then comes from comparing two analytic cores, not from extending an uncontrolled alternating convolution calculation. The Borel theorem is a separate second-saddle consequence of the coefficient equivalent.

Result	Dependencies and status
Quadratic inverse equivalent	Exact formal inversion, bare saddle, analytic twist, and positive multiple-tail merger; proved here without using the earlier forward theorem.
Sharp first omitted action	Arbitrarily accurate large fixed core, analytic core jets, and coefficient transfer of their difference; proved here.
Borderline tail constant	Same merger, uniform saddle evaluation of ε_j ; proved here.
Inverse Borel growth	Main inverse equivalent and a second real saddle; proved here.
Forward–inverse Borel ratio	The preceding transfer plus the earlier forward coefficient equivalent (8.13); this is the only inherited nontrivial asymptotic input.
Finite numerical tests	Exact checks of finite algebra and non-rigorous decimal diagnostics; not proofs of asymptotic limits.

The resolved question is Conjecture `conj:quadratic-inverse` in the pinned *ProveIt* manuscript. The finite-perturbation extension, sharp first-omitted-action law, borderline slope constant, and sharp inverse Borel equivalent are the additional proposed contributions. The classical formal inversion machinery and the general saddle-point strategy are not new. The bibliography supplies context and direct sources, but the literature search is not an exhaustive priority determination.

All asymptotics are for fixed parameters and fixed core indices. There is no claim of uniformity as $a \downarrow 0$, as a primitive parameter grows with n , or as the cutoff tends to infinity. There is no evaluation of the original divergent positive-argument feedback kernel, no all-direction resurgence result, and no proof-assistant certification.

The principal methodological point is concrete. Reversion at the quadratic boundary can be controlled by retaining small actions analytically before bounding the large-action part. A coarse many-tail estimate, strong enough to survive the remaining polynomial cancellation, is sufficient to recover both the leading inverse equivalent and the much sharper error of a small explicit core. This separates the hard global cancellation step from the local algebra which determines the first neglected interaction.

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